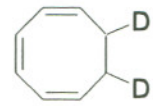
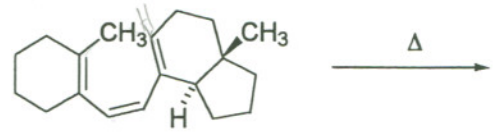
Organic Cume
October 4, 2002

For the questions given below, give detailed symmetry rules for each reaction. For questions 2 - 4, state in appropriate symmetry rule arguments, what is expected in each transformation and the effect on the stereochemistry of the reaction considered. Use orbital diagrams including stereochemistry for questions 2 and 3.

1. Show how the compound below can be used to test the allowedness of [1,3] and [1,5] sigmatropic shifts.

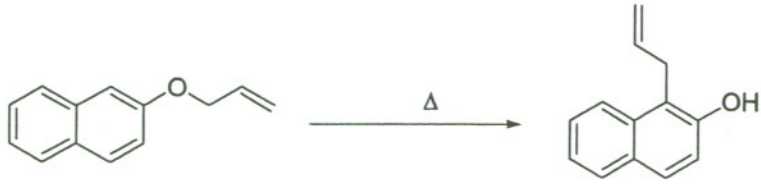


2. Predict product stereochemistry for the following reactions:

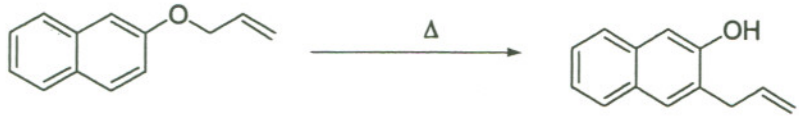


3. Predict the products and stereochemistry for thermal electrocyclic ring opening of the following two compounds. Which reacts faster?

4.



allowed or forbidden?
type of rxn?



not formed in above rxn
why?